

Lecture 5

Importance of kinetics

- Amount of product or intermediate or reactant at a given time.
- Stability and rate of drug metabolism: Dose and expiry date
- Degradation/decomposition rate.
- Slow or fast reaction.
- Radiometric dating.

Slow and Fast reactions

- This can be used to our advantage:
- The degradation of paint should be slow.
- Rusting of Iron need to be slowed down.
- Burning of cooking gas should be fast
- The rate of oxidation of a fuel should be negligible at room temperature, but it should be fast at the high temperature at which engine operates.

Importance of Kinetics

- Mechanism of reaction
- Effect of solvent on the reaction.
- Effect of substrate on the reaction.
- Effect of base/nucleophile/ electrophile on the reaction.

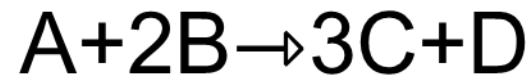
Importance of Catalysis

- Homogeneous catalysis
 - Acid-Base Catalysis
 - Enzyme Catalysis
- Heterogeneous catalysis

The rate of reaction

$$v = \frac{1}{V} \frac{d\xi}{dt} \qquad dn_i = \nu_i d\xi$$

$$v = \frac{1}{V} \frac{1}{\nu_i} \frac{dn_i}{dt} = \frac{1}{\nu_i} \frac{dC_i}{dt}$$



$$v = \frac{d[\text{D}]}{dt} = \frac{1}{3} \frac{d[\text{C}]}{dt} = -\frac{d[\text{A}]}{dt} = -\frac{1}{2} \frac{d[\text{B}]}{dt}$$

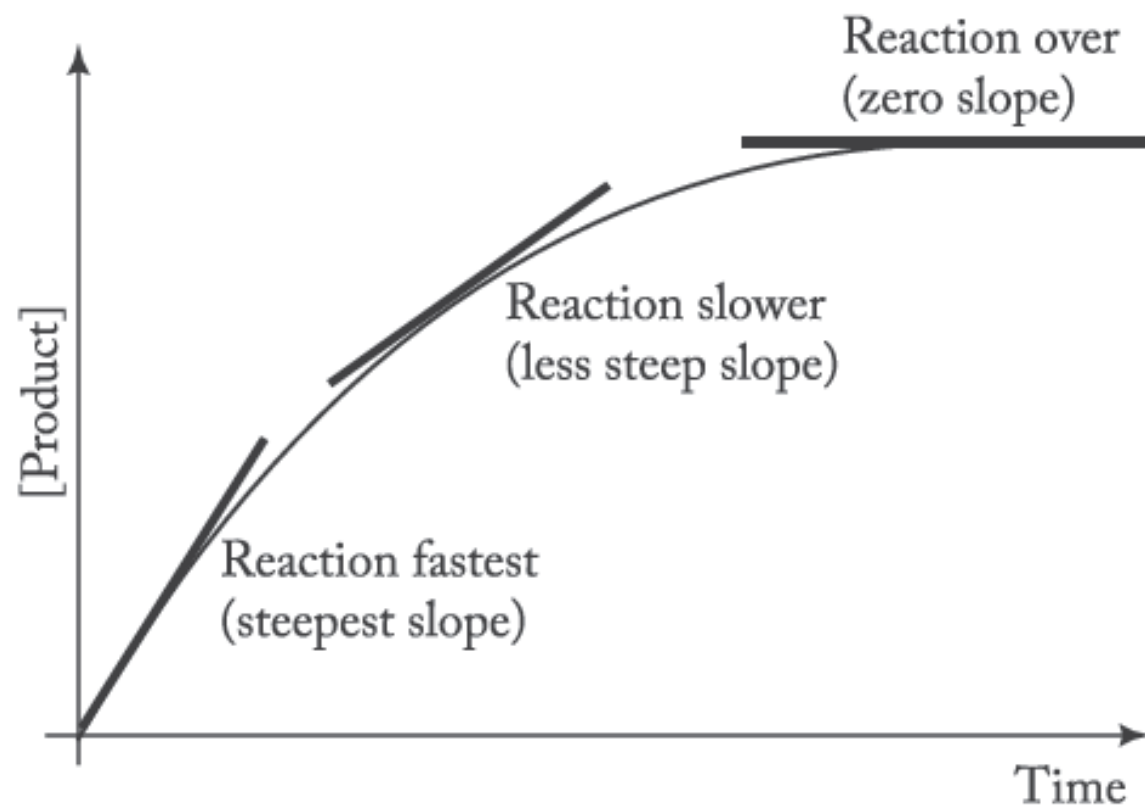


Fig. 6.1 Concentration of product as a function of time can be used to calculate the rate of reaction

Factors affecting the rate of reaction

- Concentration of reactants
- Temperature
- Mechanism of reaction:
 - Nature of reactant
 - Stability of intermediate
 - Nature of solvent
- Surface area of Reactant
- Catalyst

Rate Law

- Relation between rate of reaction and concentration of the reactants

$$\mathcal{R} = f([A], [B], \dots)$$

$$\mathcal{R} = k[A]^a[B]^b \dots$$

Molecularity: The number of molecules that come together to react in one elementary step

Three important problems

- Determination of rate law and rate constant from experimental data.
- Construct reaction mechanism that are consistent with rate law.
- Value of rate constant and their temperature dependence

Determination of rate law

- Reactions involving single reactant
 - Differential method

$$\frac{v}{[A]^n} = k$$

$$\ln v = \ln(k[A]^n)$$

$$\ln v = \ln k + n \ln [A]$$

Integrated Rate Laws

Consider a simple 1st order rxn: $A \rightarrow B$

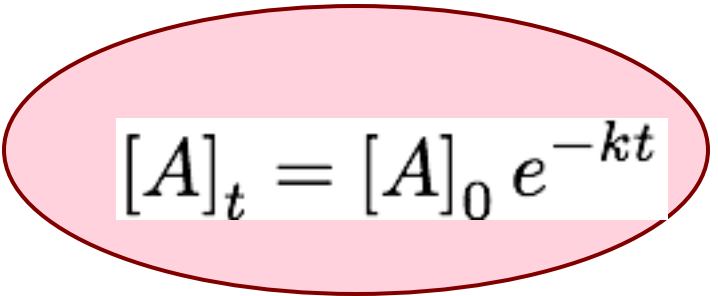
$$rate = k [A] \quad \text{Differential form:} \quad -\frac{d[A]}{dt} = k [A]$$

How much A is left after time t ? Integrate:

$$-d[A] = k [A] dt$$

$$\frac{d[A]}{[A]} = -k dt$$

$$\int \frac{d[A]}{[A]} = - \int k dt$$


$$[A]_t = [A]_0 e^{-kt}$$

Integrated Rate Laws

The integrated form of first order rate law:

$$[A]_t = [A]_0 e^{-kt}$$

Can be rearranged to give:

$$\ln \frac{[A]_t}{[A]_0} = -kt$$

$[A]_0$ is the initial concentration of A ($t=0$).

$[A]_t$ is the concentration of A at some time, t , during the course of the reaction.

Integrated Rate Laws

Manipulating this equation produces...

$$\ln \frac{[A]_t}{[A]_0} = -kt$$

$$\ln [A]_t - \ln [A]_0 = -kt$$

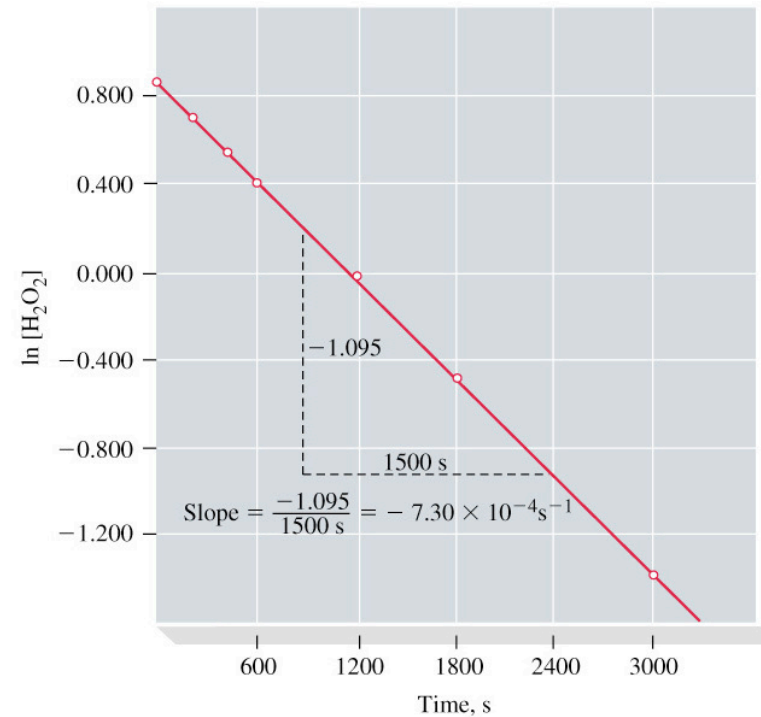
$$\ln [A]_t = -kt + \ln [A]_0$$

...which is in the form $y = mx + b$

First-Order Processes

$$\ln [A]_t = -kt + \ln [A]_0$$

If a reaction is first-order, a plot of $\ln [A]_t$ vs. t will yield a straight line with a slope of $-k$.



Second-Order Processes

Similarly, integrating the rate law for a process that is second-order in reactant A:

$$rate = -\frac{d[A]}{dt} = k[A]^2$$

Rearrange, integrate:

$$\frac{1}{[A]_t} = -kt + \frac{1}{[A]_0}$$

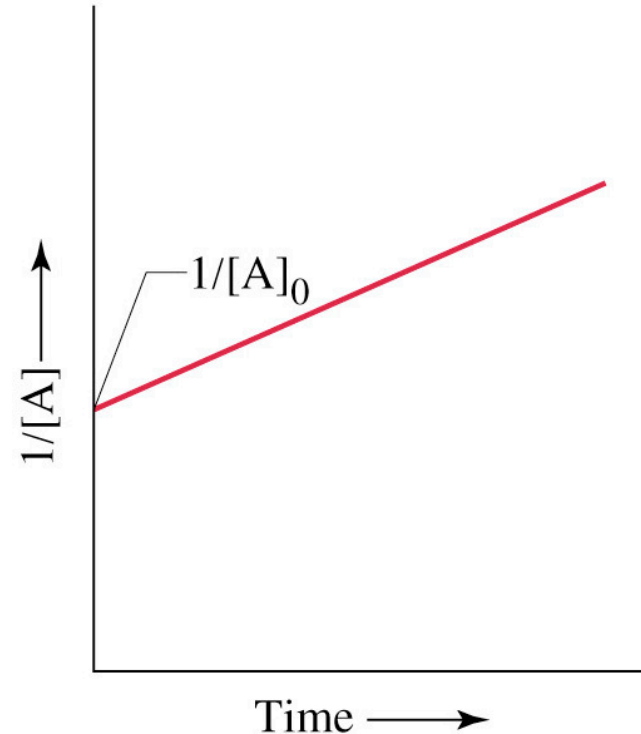
also in the form

$$y = mx + b$$

Second-Order Processes

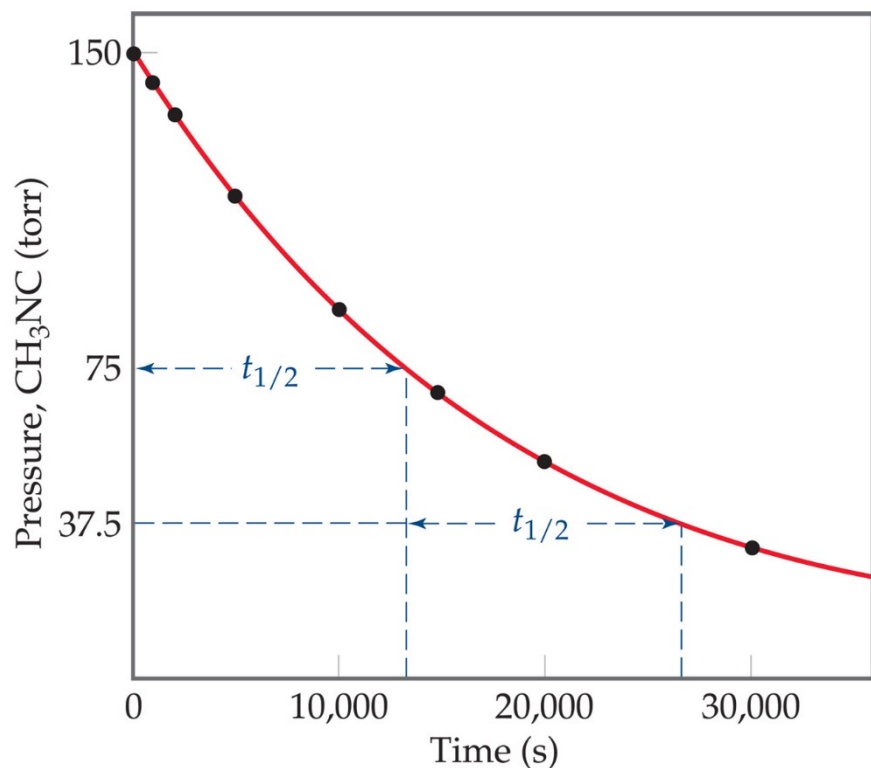
$$\frac{1}{[A]_t} = -kt + \frac{1}{[A]_0}$$

So if a process is second-order in A, a plot of $1/[A]$ vs. t will yield a straight line with a slope of k .



Order of reaction	Integrated rate equation	Linear plot
Zero order	$[A]_0 - [A] = kt$	$[A]$ vs t
First order	$\ln \frac{[A]_0}{[A]} = kt$	$-\ln[A]$ vs t
Second order	$\frac{1}{[A]} - \frac{1}{[A]_0} = kt$	$\frac{1}{[A]}$ vs t
nth order	$\frac{1}{[A]^{n-1}} - \frac{1}{[A]_0^{n-1}} = (n-1)kt$	$\frac{1}{[A]^{n-1}}$ vs t

Half-Life



- Half-life is defined as the time required for one-half of a reactant to react.
- Because $[A]$ at $t_{1/2}$ is one-half of the original $[A]$,

$$[A]_t = 0.5 [A]_0.$$

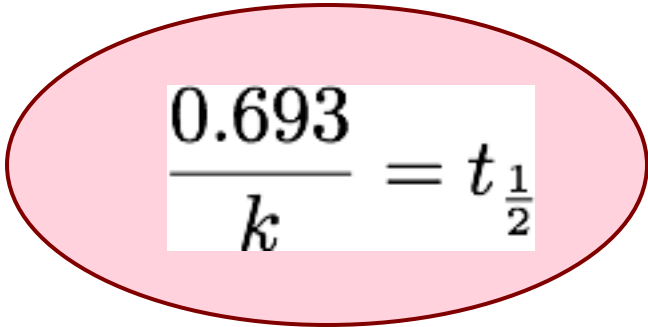
Half-Life

For a first-order process, set $[A]_t = 0.5 [A]_0$ in integrated rate equation:

$$\ln \frac{0.5 [A]_0}{[A]_0} = -kt_{\frac{1}{2}}$$

$$\ln(0.5) = -kt_{\frac{1}{2}}$$

$$\ln(2) = 0.693 = -kt_{\frac{1}{2}}$$


$$\frac{0.693}{k} = t_{\frac{1}{2}}$$

NOTE: For a first-order process, the half-life does not depend on $[A]_0$.

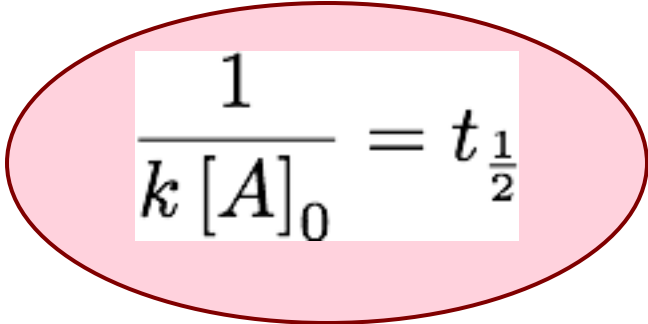
Half-Life- 2nd order

For a second-order process, set
 $[A]_t = 0.5 [A]_0$ in 2nd order equation.

$$\frac{1}{0.5 [A]_0} = kt_{\frac{1}{2}} + \frac{1}{[A]_0}$$

$$\frac{2}{[A]_0} = kt_{\frac{1}{2}} + \frac{1}{[A]_0}$$

$$\frac{2}{[A]_0} - \frac{1}{[A]_0} = kt_{\frac{1}{2}}$$


$$\frac{1}{k [A]_0} = t_{\frac{1}{2}}$$

Half-Life

Concentration dependence of half-life	Order of reaction
$t_{\frac{1}{2}} \propto [A]_0$	Zero order
$t_{\frac{1}{2}} \propto ([A]_0)^0$	First order
$t_{\frac{1}{2}} \propto ([A]_0)^{-1}$	Second order
$\succ t_{\frac{1}{2}} \propto ([A]_0)^{n-1}$	nth order